

Authentication Logs — Cornell Deep

Real logs exported from `data/authshield360.db` (table `audit_trail`) after running the four test suites. Every attempt — success or failure — is recorded exactly once via `commitAudit()`.

1) Summary

Item	Value
Total recorded attempts	41
Rows shown below	41 (newest first)
Average risk score	40.7
Highest risk score	100
Non-successful outcomes	28
Active sessions	12
Trusted devices	10

2) Distribution by outcome

Outcome	Count	Avg risk	Avg time (ms)
SUCCESS	13	10.4	8.8
HIGH_RISK_BLOCK	9	100	3.4
FAILED_PASSWORD	8	55	8.6
STEP_UP	6	32.5	4.8
INVALID_OTP	2	0	3
RECOVERY_RESET	1	0	null
RECOVERY_REQUEST	1	0	null
RECOVERY_CODE_VALID	1	0	null

3) By scenario (1 = password · 2 = + OTP · 3 = + e-mail and RBAC)

Scenario	Attempts	Success	Avg time (ms)
1	18	4	6
2	13	4	5
3	5	3	12

4) Top IP addresses by rejections

IP address	Network	Attempts	Failures
127.0.0.1	Localhost Loopback	18	12
10.10.4.25	Corporate Campus LAN	10	6
185.220.101.4	Public Internet Edge	5	5
10.0.8.77	Corporate Campus LAN	2	1
45.55.200.100	Public Internet Edge	1	1
203.0.113.77	Public Internet Edge	1	1

5) Most frequent risk reasons

Outcome	Risk factors	Frequency
SUCCESS	BASELINE	7

Outcome	Risk factors	Frequency
HIGH_RISK_BLOCK	ACCOUNT_LOCKED	6
FAILED_PASSWORD	BASELINE	3
STEP_UP	CROSS_DEVICE_PENDING, NEW_IP, UNRECOGNIZED_DEVICE	3
FAILED_PASSWORD	NEW_IP, UNRECOGNIZED_DEVICE, CONSECUTIVE_FAILURES	2
FAILED_PASSWORD	NEW_IP, UNRECOGNIZED_DEVICE	2
HIGH_RISK_BLOCK	IP_BLACKLISTED	2
INVALID_OTP	BASELINE, OTP	2
STEP_UP	BASELINE	2
SUCCESS	NEW_IP	2

6) Detailed log (last 60 attempts)

Time	User	Role	Outcome	Risk	Risk factors	IP	Device	Scenario	Time (ms)
2026-09-28T18:42:30.678Z	admin	ADMINISTRATOR	SUCCESS	30	UNRECOGNIZED_DEVICE	127.0.0.1	Edge · Campus PC	3	10
2026-09-28T18:42:25.571Z	demo1790620914677	STUDENT	STEP_UP	30	UNRECOGNIZED_DEVICE	127.0.0.1	Unknown Device	2	1
2026-09-28T18:42:24.197Z	rec1790620939694	STUDENT	FAILED_PASSWORD	20	BASELINE	127.0.0.1	Unknown Device	1	9
2026-09-28T18:42:24.165Z	rec1790620939694	STUDENT	SUCCESS	0	BASELINE	127.0.0.1	Unknown Device	1	9
2026-09-28T18:42:24.130Z	rec1790620939694	STUDENT	RECOVERY_CODE_VALID	0	—	127.0.0.1	DEV-71f8e7 · Unknown Device	—	—
2026-09-28T18:42:24.097Z	rec1790620939694	STUDENT	RECOVERY_RESET	0	MFA_RESET	127.0.0.1	DEV-6a7078 · Unknown Device	—	—
2026-09-28T18:42:23.955Z	rec1790620939694	STUDENT	RECOVERY_REQUEST	0	—	127.0.0.1	DEV-6a7078 · Unknown Device	—	—
2026-09-28T18:42:23.897Z	rec1790620939694	STUDENT	SUCCESS	25	NEW_IP	127.0.0.1	Unknown Device	2	9
2026-09-28T18:42:17.454Z	teacher2	TEACHER	HIGH_RISK_BLOCK	100	IP_BLACKLISTED	185.220.101.4	DEV-813221 · sim	1	1
2026-09-28T18:42:15.387Z	teacher2	TEACHER	HIGH_RISK_BLOCK	100	ACCOUNT_LOCKED	203.0.113.10	DEV-813221 · sim	2	1
2026-09-28T18:42:14.150Z	teacher2	TEACHER	HIGH_RISK_BLOCK	100	ACCOUNT_LOCKED	45.55.200.100	DEV-813221 · sim	2	2
2026-09-28T18:42:12.186Z	teacher2	TEACHER	HIGH_RISK_BLOCK	100	ACCOUNT_LOCKED	185.220.101.4	DEV-813221 · sim	1	1
2026-09-28T18:42:12.149Z	teacher2	TEACHER	HIGH_RISK_BLOCK	100	ACCOUNT_LOCKED	185.220.101.4	DEV-813221 · sim	1	1
2026-09-28T18:42:12.104Z	teacher2	TEACHER	FAILED_PASSWORD	95	NEW_IP, UNRECOGNIZED_DEVICE, CONSECUTIVE_FAILURES	185.220.101.4	DEV-813221 · sim	1	9
2026-09-28T18:42:12.076Z	teacher2	TEACHER	FAILED_PASSWORD	75	NEW_IP, UNRECOGNIZED_DEVICE	185.220.101.4	DEV-813221 · sim	1	11
2026-09-28T18:42:11.957Z	student1	STUDENT	HIGH_RISK_BLOCK	100	IP_BLACKLISTED	203.0.113.77	Unknown Device	1	1
2026-09-28T18:42:11.876Z	admin	ADMINISTRATOR	SUCCESS	0	TRUSTED_DEVICE_APPROVAL	127.0.0.1	Unknown Device	—	—

Time	User	Role	Outcome	Risk	Risk factors	IP	Device	Scenario	Time (ms)
2026-09-28T18:42:11.828Z	admin	ADMINISTRATOR	STEP_UP	55	CROSS_DEVICE_PENDING, NEW_IP, UNRECOGNIZED_DEVICE	127.0.0.1	Unknown Device	3	10
2026-09-28T18:42:11.766Z	student2	STUDENT	HIGH_RISK_BLOCK	100	ACCOUNT_LOCKED	127.0.0.1	Unknown Device	1	1
2026-09-28T18:42:11.725Z	student2	STUDENT	FAILED_PASSWORD	95	NEW_IP, UNRECOGNIZED_DEVICE, CONSECUTIVE_FAILURES	127.0.0.1	Unknown Device	1	8
2026-09-28T18:42:11.701Z	student2	STUDENT	FAILED_PASSWORD	75	NEW_IP, UNRECOGNIZED_DEVICE	127.0.0.1	Unknown Device	1	11
2026-09-28T18:42:11.624Z	demo1790620914677	STUDENT	STEP_UP	55	CROSS_DEVICE_PENDING, NEW_IP, UNRECOGNIZED_DEVICE	172.16.0.55	Unknown Device	2	8
2026-09-28T18:42:11.579Z	demo1790620914677	STUDENT	SUCCESS	0	TRUSTED_DEVICE_APPROVAL	10.0.8.77	Unknown Device	—	—
2026-09-28T18:42:11.527Z	demo1790620914677	STUDENT	STEP_UP	55	CROSS_DEVICE_PENDING, NEW_IP, UNRECOGNIZED_DEVICE	10.0.8.77	Unknown Device	2	9
2026-09-28T18:42:11.480Z	teacher2	TEACHER	SUCCESS	0	BASELINE	10.20.0.9	Unknown Device	3	9
2026-09-28T18:42:11.433Z	teacher2	TEACHER	SUCCESS	55	NEW_IP, UNRECOGNIZED_DEVICE	10.20.0.9	Unknown Device	2	11
2026-09-28T18:42:11.401Z	demo1790620914677	STUDENT	INVALID_OTP	0	BASELINE, OTP	127.0.0.1	Unknown Device	2	3
2026-09-28T18:42:11.344Z	rbac1790620927356	STUDENT	HIGH_RISK_BLOCK	100	UNAUTHORIZED_ROUTE	127.0.0.1	Unknown Device	3	22
2026-09-28T18:42:07.357Z	demo1790620914677	STUDENT	STEP_UP	0	BASELINE	127.0.0.1	Unknown Device	2	0
2026-09-28T18:42:07.337Z	demo1790620914677	STUDENT	SUCCESS	0	BASELINE	127.0.0.1	Unknown Device	1	7
2026-09-28T18:42:07.306Z	demo1790620914677	STUDENT	SUCCESS	25	NEW_IP	127.0.0.1	Unknown Device	2	7
2026-09-28T18:41:54.393Z	new.student1790620908897	STUDENT	SUCCESS	0	BASELINE	10.10.4.25	smoke	1	13
2026-09-28T18:41:48.871Z	student2	STUDENT	HIGH_RISK_BLOCK	100	ACCOUNT_LOCKED	10.10.4.25	DEV-6a7078 · Unknown Device	1	1
2026-09-28T18:41:48.841Z	student2	STUDENT	FAILED_PASSWORD	40	CONSECUTIVE_FAILURES	10.10.4.25	DEV-6a7078 · Unknown Device	1	8
2026-09-28T18:41:48.829Z	student2	STUDENT	FAILED_PASSWORD	20	BASELINE	10.10.4.25	DEV-6a7078 · Unknown Device	1	6
2026-09-28T18:41:48.794Z	admin	ADMINISTRATOR	SUCCESS	0	BASELINE	10.10.4.25	DEV-6a7078 · Unknown Device	3	7
2026-09-28T18:41:48.769Z	teacher1	TEACHER	INVALID_OTP	0	BASELINE, OTP	10.10.4.25	DEV-6a7078 · Unknown Device	2	3
2026-09-28T18:41:48.748Z	teacher1	TEACHER	SUCCESS	0	BASELINE	10.10.4.25	DEV-6a7078 · Unknown Device	2	7
2026-09-28T18:41:48.742Z	teacher1	TEACHER	STEP_UP	0	BASELINE	10.10.4.25	DEV-6a7078 · Unknown Device	2	1
2026-09-28T18:41:48.720Z	student1	STUDENT	FAILED_PASSWORD	20	BASELINE	10.10.4.25	DEV-6a7078 ·	1	7

Time	User	Role	Outcome	Risk	Risk factors	IP	Device	Scenario	Time (ms)
							Unknown Device		
2026-09-28T18:41:48.698Z	student1	STUDENT	SUCCESS	0	BASELINE	10.10.4.25	DEV-6a7078 · Unknown Device	1	8

7) Brute-force protection counters

Metric	Value	Meaning
Failed password (FAILED_PASSWORD)	8	counts toward CONSECUTIVE_FAILURE (weight 20)
Invalid OTP (INVALID_OTP)	2	wrong or expired code
Medium step-up (STEP_UP)	6	an OTP/e-mail code was required before allowing access
High-risk block (HIGH_RISK_BLOCK)	9	risk ≥ 70, blacklisted IP, or RBAC breach
Success (SUCCESS)	13	also written to success_logins

Every counter above is updated in the same moment the audit row is written, so what the SOC dashboard shows is exactly what is stored — which is what `restart-check.js` (18 checks) verifies after restarting the process.